Images of wild rice diseases

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Fungal Brown Spot caused by the fungus (*Bipolaris oryzae* = *Cochliobolus miyabeanus*)

Field images



Greenhouse after inoculation with *B. oryzae*

2 days



7 days



30 days



Note: This is the most important disease. Starts like small brownish to purple spots and then enlarge, mostly elongate and could develop chlorosis. Lesions can coalescence as well. There is a lot of variation in symptoms in the field depending of the growth stage of wild rice during the observations and likely the effectors of the fungus.

Spot Blotch caused by the fungus (*Bipolaris sorokiniana* = *Cochliobolus sativus*)

Greenhouse symptoms after inoculation

Fully developed lesions











Lesions in an older leaf (note the orange redness color in the edges and tip due to senescence)

Note: These images are from inoculation in GH. Please note that they shape of the lesions are more oval shape than those of fungal brown spot, and can develop chlorosis; they also coalescence and can become tan (dry). There is variation in symptoms shape in GH and field. T:he fungus is frequently found in paddies.

Eyespot caused by the fungus (*Bipolaris gigantea* syn=*Drechslera gigantea*)

Field images

Early symptoms





Mature symptoms

Note: Please note that lesions appear as water soak areas (first image) and then they look like “eyes” due to secondary infections when lesions mature. The fungus is found in paddies but the occurrence vary year by year

Physiological lesions: “Johnson lesions

Field images





These lesions are expressed in the wild rice germplasm called Johnson. It is believed that they can be spontaneous, or/and induced/enhanced by fungi. We are starting working on genetic mechanisms.

Leaf blotch putatively caused by Phaeoseptoria spp.

Field image



Mature lesion (it can develop a black (necrotic) edge)

Other diseases include:

- Leaf sheath and stem rot (*Nakatea sigmoidea* and *Ceratorhiza hydrophila*)
- Scab (*Fusarium* spp.)
- Smut (*Entyloma lineatum*)
- Bacteria brown spot (*Pseudomonas syringea* pv. nov. *zizania*)
- Bacteria Leaf Streak (*Xanthomona campestris*)